



MAHARAJ SHREE KUMAR SOOD

CONTACT

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LINKS

- [My LinkedIn Profile](#)

EDUCATION

September 2014

Master of Technology:

Embedded Systems

Amity University, Noida

May 2012

Bachelor of Technology:

Electronics & Communication

MRCE, Faridabad

CERTIFICATIONS

	CERTIFIED IN CYBERSECURITY
	ISO 27001 LEAD AUDITOR
	CERTIFIED 62443 PROFESSIONAL
	CISCO CERTIFIED CYBERSECURITY ESSENTIALS

PROFESSIONAL SUMMARY

Senior Cybersecurity Professional with **12+ years of experience** across **Automotive Cybersecurity, ICS/OT Security, Embedded Systems, and Connected Platforms**.

Experienced in delivering **security-by-design architectures, cybersecurity risk assessments, and regulatory compliance programs** aligned with **ISO/SAE 21434, UNECE R155/R156, IEC 62443, and NIST frameworks**.

Proven ability to lead **vehicle cybersecurity engineering initiatives, CSMS/SUMS framework implementation, and OT security programs** while collaborating with OEMs, engineering teams, and stakeholders to enhance **product security, regulatory compliance, and operational resilience**.

SKILLS

Cybersecurity Domains

- Automotive Cybersecurity
- ICS / OT Security
- IoT & Embedded Security
- Secure SDLC & Security-by-Design

Automotive Systems

- Connected ECUs (IVI, Cluster, Telematics)
- Body Control Modules (BCM), Secure Gateways
- HPC & Zonal ECUs
- Powertrain Systems (ICE & EV)

Security Standards & Regulations

- ISO/SAE 21434
- UNECE WP.29 (R155, R156)
- IEC 62443, NERC CIP, NCIIPC
- ISO 27001, NIST Cybersecurity Framework
- RED, CRA

Industrial Systems

- SCADA, DCS, PLC Security
- Industrial Network Security
- OT Risk Assessments
- Component Cyber Compliance
- Critical Infra, Oil and Gas

Security Technologies

- PKI, TLS, Cryptography
- Secure Boot, Debug, Interface, Firmware Update
- IDS/IPS, Firewalls
- Vulnerability Assessment & Penetration Testing

	IBM CERTIFIED CYBERSECURITY ANALYST
	OMNEX CERTIFIED 21434 AUTOMOTIVE CYBER PROFESSIONAL
	CERTIFIED NIST CSF LEAD IMPLEMENTER
	Cert-In Certified Cyber Crisis Management Professional (CCMP)

ACHIEVEMENTS

- Awarded **Silver Medal** for securing **2nd rank in Master of Technology (Embedded Systems)**.
- **Recognised by the Indian Army for successful delivery and deployment of robotic solutions in real-time operational environments**

WORK HISTORY

April 2022 - Current

Lead Cyber Consultant, *Thales*, Noida, India

➤ **Cybersecurity Engineering, TARA & Regulatory Compliance**

- **Implement** security-by-design across the product lifecycle for connected ECUs, delivering **ISO/SAE 21434 work products** from concept through post-development.
- Lead **vehicle-level cybersecurity risk assessments and TARA workshops** across 2 OEM programs, driving security decisions with engineering and program teams.
- **Drive UNECE WP.29 R155/R156** compliance for OEM programs, preparing and validating the cybersecurity work products required for type approval submission.
- **Establish CSMS and SUMS frameworks** for automotive OEMs, achieving regulatory readiness and supporting successful type approval.
- Perform OT/ICS risk assessments and vulnerability analysis aligned to IEC 62443-3-2, 4-1 and 4-2, strengthening industrial **cybersecurity posture** across the power sector.
- **Author** cybersecurity policies, procedures, and security architecture guidelines aligned to IEC 62443 and NIST CSF, adopted as reference standards across the power sector.

➤ **Security Lab Governance, Procurement & ISMS Assurance**

- **Own** end-to-end procurement of penetration testing lab infrastructure, **from technical specification through vendor selection and commissioning**.
- Established **lab security policies and procedures** and drove the **ISO 27001 assessment** for the penetration testing lab.

April 2021 - April 2022

Technical Lead, *HCL Technologies*, Noida, India

➤ **Connected Vehicle Security Analysis & Requirements Definition**

- **Led** TARA activities **for connected vehicle platforms and infotainment systems**.
- **Assessed** FOTA architecture and Android connected applications; **designed and validated** secure boot and firmware update mechanisms **for ECU software integrity**.
- **Defined** cybersecurity requirements and mitigation strategies **aligned to NIST and OWASP**.

September 2018 - March 2021

Senior Security Developer, Marelli India Pvt Ltd.,
Manesar, India

➤ **Secure Vehicle Functionality Implementation & SW Architecture**

- Developed **secure embedded software for Body Control Modules (BCM) and instrument cluster ECUs** for global OEM programs.
- Implemented **secure remote keyless entry systems**, ensuring confidentiality and integrity of authentication frames.
- Designed vehicle **anti-theft and immobiliser security mechanisms**, strengthening overall vehicle protection.
- Implemented **secure coding practices aligned with MISRA and CERT-C standards** to reduce software vulnerabilities.
- Created **software architecture models using Enterprise Architect**, enabling scalable and secure ECU design.

February 2015 - September 2018

Embedded Research Engineer, The Hi-tech Robotics Systemz Ltd, Gurgaon, India

➤ **Teleoperated robots for bomb disposal, pick-and-place, and long-range surveillance.**

- Developed embedded software for **robotic surveillance and defence systems** supporting IED detection and remote operations.
- Designed and implemented **CAN-based communication architectures** for multi-module robotic platforms.
- Optimised system design by transitioning from a **multi-MCU architecture to a centralised controller architecture**.
- Integrated hardware interfaces including **SPI, UART, GPS, and CAN Open protocols**.
- Performed motor control tuning, system integration, and load testing to ensure robust field performance.

September 2014 - December 2014

Technical Executive, Milagrow Knowledge & Business Solutions, Gurgaon, India

➤ **Troubleshooting of Home Cleaning Robot Solutions**